1 Definition and Examples

It is a general principle in mathematics that when one defines an interesting mathematical structure, one should also study the "structure-preserving" maps (functions) between two such structures. In the case of vector spaces, the fundamental operations are addition and scalar multiplication, or, equivalently, forming linear combinations. A structure-preserving map between vector spaces is therefore one which preserves linear combinations. These are called *linear maps*.

Definition 1.1 (Linear map)**.** Let V and W be vector spaces over F. A *linear map* from V to W is a function $T : V \to W$ with the property that

$$T(\alpha v_1 + v_2) = \alpha T(v_1) + T(v_2). \tag{1.2}$$

Remark 1.3.

(a) The terms *linear function* or *linear transformation* are often used interchangeably with *linear map*. These all have exactly the same meaning. If $W = V$, then a linear map $T : V \to V$ is said to be a *linear operator* on V.

(b) If T is a linear map and v is a vector, it is common to write Tv instead of $T(v)$. These two notations also have exactly the same meaning.

Exercise 1.4. Show that the condition in Equation (1.2) is equivalent to the two conditions

$$\begin{aligned} T(v_1 + v_2) &= Tv_1 + Tv_2 \\ T(\alpha v) &= \alpha Tv \end{aligned}$$

for all $v, v_1, v_2 \in V$ and $\alpha \in F$. Some books use these two conditions to define a linear map instead of (1.2). Other books define a linear map to be one which satisfies

$$T(\alpha_1 v_1 + \alpha_2 v_2) = \alpha_1 Tv_1 + \alpha_2 Tv_2$$

for all $v_1, v_2 \in V$ and all $\alpha_1, \alpha_2 \in F$. It is easily verified that this condition is also equivalent to the other two.

Example 1.5. Take $W = V$ and consider the identity map $\mathrm{id}_V : V \to V$, defined by $\mathrm{id}_V(v) = v$ for every $v \in V$. If $v_1, v_2 \in V$ and $\alpha \in F$, then

$$\begin{aligned} \mathrm{id}_V(\alpha v_1 + v_2) &= \alpha v_1 + v_2 \\ &= \alpha \mathrm{id}_V v_1 + \mathrm{id}_V v_2, \end{aligned}$$

which shows that id_V is linear.

Exercise 1.6. Let $0 : V \to W$ denote the *zero map*, defined by $0v = 0$ for all $v \in V$. Show that the zero map is linear. (Note that 0 on the left-hand-side of this equation denotes the zero map, while 0 on the right-hand-side denotes the zero vector of the vector space W. It is common to use the same notation with different meanings when the context makes clear what each use is supposed to mean.)

Example 1.7. Let $\mathscr{P}(\mathbb{R})$ denote the set of all polynomials with real coefficients. Since the derivative of a polynomial is another polynomial, we can view differentiation as a map $D : \mathscr{P}(\mathbb{R}) \to \mathscr{P}(\mathbb{R})$. Two of the basic properties of differentiation say that $(f+g)' = f' + g'$ and $(\alpha f)' = \alpha f'$ show is that D is a linear operator on $\mathscr{P}(\mathbb{R})$.

Exercise 1.8. Define $T : \mathscr{P}(\mathbb{R}) \to \mathbb{R}$ by

$$Tp = \int_0^1 p(x)dx.$$

Show that T is a linear map.

Exercise 1.9. Define $T : \mathscr{P}(\mathbb{R}) \to \mathscr{P}(\mathbb{R})$ by $(Tp)(x) = x^2 p(x)$ for each $x \in \mathbb{R}$. Show that T is a linear map.

Exercise 1.10. Let F^{ω} denote the vector space of all sequences of elements of F. Define $T : F^{\omega} \to F^{\omega}$ by $T(x_1, x_2, x_3, \dots) = (x_2, x_3, \dots)$. Show that T is a linear operator on F^{ω}.

Exercise 1.11. Define $T : \mathbb{R}^3 \to \mathbb{R}^2$ by

$$T(x, y, z) = (2x - y + 3z, 7x + 5y - 6z).$$

Prove that T is a linear map.

Exercise 1.12. Fix a polynomial $q \in \mathscr{P}(\mathbb{R})$. Define $T : \mathscr{P}(\mathbb{R}) \to \mathscr{P}(\mathbb{R})$ by $(Tp)(x) = p(q(x))$ for each $x \in \mathbb{R}$. Show that T is a linear operator on $\mathscr{P}(\mathbb{R})$.

2 Linear Map Lemma

Lemma 2.1 (Linear Map Lemma)**.** *Let V be a finite-dimensional vector space and W a vector space. Suppose $\{v_1,\dots,v_n\}$ is a basis of V and $w_1,\dots,w_n$ any n vectors in W. Then there exists a unique linear map $T : V \to W$ such that*

$$Tv_k = w_k$$

for each $k = 1,\dots,n$.

Remark 2.2. The existence part of the Linear Map Lemma means that we can find a linear map that maps a basis to any vectors we wish. The uniqueness part says that a linear map is completely determined by its values on a basis.

Exercise 2.3. Prove Lemma (2.1). (*Hint*: Show that $Tv = \alpha_1 w_1 + \cdots + \alpha_n w_n$. Explain how to choose the scalars α_i, why this uniquely defines T, and show that T is linear.)

Proposition 2.4. *If $T : V \to W$ is a linear map, then $T(0) = 0$. (*Hint*: $T(0) = T(0+0)$)*

Exercise 2.5. Prove Proposition 2.4.

Exercise 2.6. Fix $m, b \in \mathbb{R}$. Show that the function $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = mx + b$ is a linear map if and only if $b = 0$. This shows that the linear functions of basic algebra are not the same as linear maps in the context of linear algebra.

Exercise 2.7. Let $M^{m\times n}(F)$ be the vector space of $m \times n$ matrices with entries in F. Given a matrix $A \in M^{m\times n}(F)$, we can define a map $T : M^{m\times 1}(F) \to M^{n\times 1}(F)$ by $T(x) = Ax$ by matrix multiplication. Show that this is linear. (*Hint*: Write this in components: the $T(x)_i = \sum_{k=1}^{n} A_{ij}x_j$.)

3 Algebra of Linear Maps

Fix vector spaces V and W over F. Let $\mathscr{L}(V,W)$ denote the set of all linear maps $T : V \to W$. In the case $W = V$ we write $\mathscr{L}(V) := \mathscr{L}(V,V)$. Given $S,T \in \mathscr{L}(V,W)$ and $\alpha \in F$, we define the sum $S+T$ and scalar multiple αT by

$$(S+T)(v) := Sv + Tv,$$
$$(\alpha T)(v) := \alpha Tv.$$

Remark 3.1. Another common notation for $\mathscr{L}(V,W)$ which you may encounter elsewhere is $\mathrm{Hom}(V,W)$. This is a more general notation which comes from category theory, emphasizing that linear maps are the morphisms in the category of vector spaces over F.

Exercise 3.2. Prove that $\mathscr{L}(V,W)$ is a subspace of W^V, the vector space of *all* maps from $V \to W$. Note that we showed last time that the zero map is linear, so $0 \in \mathscr{L}(V,W)$. All that remains is to shows that $\mathscr{L}(V,W)$ is closed under the operations above (or, equivalently, under linear combinations).

It usually makes no sense to multiply two elements of a vector space, but for some pairs of linear maps a useful product does exist.

Definition 3.3 (Product of Linear Maps)**.** If $T \in \mathscr{L}(U,V)$ and $S \in \mathscr{L}(V,W)$, then we can define the *product* $ST \in \mathscr{L}(U,W)$ by $ST := S \circ T$. That is, for each $u \in U$,

$$(ST)(u) = S(Tu)).$$

Note that, just as for ordinary functions, ST is only defined when the range of T is contained in the domain of S.

Exercise 3.4. If $T \in \mathscr{L}(U,V)$ and $S \in \mathscr{L}(V,W)$, prove that $ST \in \mathscr{L}(U,W)$. That is, prove that ST is linear.

Exercise 3.5. Let $D \in \mathscr{L}(\mathscr{P}(\mathbb{R}))$ be differentiation and $T \in \mathscr{L}(\mathscr{P}(\mathbb{R}))$ be multiplication by x^2. We showed last time that both of these are linear maps. Show that $TD \neq DT$. This shows that the product of linear maps, in general, is not commutative.

In the case of $\mathscr{L}(V)$, the product of two linear operators is always defined. It is straightforward to verify that the product of linear operators on V has the following properties:

(a) (Bilinearity) For all $S, T_1, T_2 \in \mathscr{L}(V)$ and all $\alpha \in F$,

- $S(T_1 + T_2) = \alpha ST_1 + ST_2$
- $(S_1 + S_2)T = S_1T + S_2T$
- $(\alpha S)T = \alpha(ST) = S(\alpha T)$

(b) (Associativity) $(RS)T = R(ST)$ for all $R, S, T \in \mathscr{L}(V)$

(c) (Identity) $IT = TI = T$ for all $T \in \mathscr{L}(V)$, where I is the identity map.

These properties are said to make $\mathscr{L}(V)$ into an *unital associative F-algebra*.